

Year 2 Quantum Mechanics - Example Sheet 3

All of the material for this sheet has already been covered.

1. The energies of states describing particles bound in a 3-dimensional box of side a are given by:

$$E = \frac{\pi^2 \hbar^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2),$$

where n_x , n_y and n_z are integers.

- a) What is the lowest possible value for the energy E ? What is the degeneracy corresponding to this value?
 - b) What is the second lowest value of the energy E ? What is the degeneracy corresponding to this value? Given the probability function $|\psi|^2$ for each of these states, could they be distinguished from each other?
 - c) Let $n^2 = n_x^2 + n_y^2 + n_z^2$ be an integer proportional to a given permitted energy. List the degenerate solutions corresponding to $n^2 = 3, 6, 9, 11$ and 12 . Can you find values of n^2 for which the energy level is six-fold degenerate?
2. A particle in an infinite square well extending from $x = 0$ to $x = L$ has the normalised wavefunction:

$$\Psi(x, t) = A \left(2 \sin\left(\frac{\pi x}{L}\right) e^{-iE_1 t/\hbar} + \sin\left(\frac{2\pi x}{L}\right) e^{-iE_2 t/\hbar} \right),$$

where $A = \sqrt{2/5L}$ and $E_n = n^2 \pi^2 \hbar^2 / 2mL^2$.

Calculate the probability density $P(x, t)$ and show that it oscillates in time with an angular frequency $\omega = \frac{(E_2 - E_1)}{\hbar}$. Show that the value of $P(x, t)$ at $x = \frac{L}{2}$ is a constant. Show that:

$$\langle p_x \rangle = \frac{32\hbar}{15L} \sin(\omega t).$$

Sketch the time dependence of $\langle p_x \rangle$ and describe what is happening.

You may need $\int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{L}\right) dx = 0$, $\int_0^L \sin\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi x}{L}\right) dx = 0$ and $\int_0^L \sin\left(\frac{2\pi x}{L}\right) \cos\left(\frac{2\pi x}{L}\right) dx = 0$.

3. The atoms in a volume of gas are excited at time $t = 0$ by a short electrical discharge. On de-exciting the atoms emit light of wavelength $6 \times 10^{-7} \text{ m}$, corresponding to a unique transition, with an intensity given by:

$$I(t) = I_0 e^{-\gamma t}.$$

If the value of the decay constant γ is 10^8 s^{-1} calculate the spread in the wavelength of the light. This is the natural width of the spectral line.